

Rohit Singh

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SUMMARY

Innovative Senior Software Engineer with 4.11 years of experience in AI-driven enhancements and large language models. Developed SentimentIQ leveraging GPT-5, achieving an 85% reduction in manual feedback analysis and real-time processing of over 10,000 responses. Seeking to elevate technological solutions by integrating advanced prompt engineering techniques and retrieval-augmented generation in future projects, streamlining processes and driving efficiency through cutting-edge AI methodologies.

EXPERIENCE

- **Tech Stacks:** Python, Flask, OpenAI GPT-5, Claude AI, Prompt Engineering, Few-Shot Learning, RAG (Retrieval-Augmented Generation), MongoDB Atlas Vector Search, Semantic Search, REST APIs, Authentication & Session Management, AI/LLM Integration, Chat Analytics, Vector Embeddings, NLP, Git, Cloud Deployment
- Engineered advanced prompt engineering strategies for OpenAI GPT-5 and Claude to achieve contextual understanding of customer feedback.
- Implemented RAG architecture using MongoDB Atlas vector embeddings for semantic similarity search.
- Built scalable Flask API with real-time session management system with user authentication and personalized chat history tracking.
- Utilized OpenAI and few-shot prompt engineering to classify and analyse over 20,000 customer chats per month, enabling the identification of key drivers behind customer sentiments through automated processes.
- Designed a feedback triage pipeline that routes clustered sentiment themes to product owners, turning unstructured chat logs into prioritized action items for faster issue resolution.

Senior Software Engineer

Optum Global Solutions

March 2024 - Present, Noida, India

- Developed SentimentIQ using advanced prompt engineering techniques with OpenAI GPT-5 and Claude LLMs, implementing RAG architecture and MongoDB vector embeddings reducing manual feedback analysis by 85% and enabling real-time processing of 10,000+ customer responses into instant business insights that gives direct actionable items.
- Developed a search service transforming data from technical websites, files, and SharePoint into a unified semantic search platform using ML techniques and a Cosmos vector database. Indexed over 10,000 files with search times under 5 seconds, saving time for developers and leadership while reducing operational costs.
- Developed a Chatbot using Retrieval-Augmented Generation and LangChain to integrate OpenAI services, converting data into embeddings and storing in Cosmos DB. Enhanced search response speed by up to 80%, significantly improving user interaction efficiency and enabling faster, more accurate query resolution.
- Developed a tool recommendation LangChain, LangChain OpenAI, to provide the best AI tool for the problem statement.

Software Engineer

Optum Global Solutions

June 2021 – February 2024, Bangalore, India

- Developed an automated reporting tool for GitHub repositories using Python, GitHub APIs, and GraphQL, improving visibility across 100,000+ repositories and eliminating manual tracking. Enabled quick identification of security vulnerabilities, boosting efficiency and security monitoring.
- Migrated schema and data from TigerGraph (Graph DB) to MySQL (Relational DB), saving \$10k per month in licensing costs. Transitioned multiple services to use Spring Data for more efficient database operations, improving performance and scalability.
- Implemented Microsoft Power Automate to record and monitor usage, and to survey user satisfaction in an automated manner, reducing operational costs and minimizing the need for manual user interactions.
- Established a centralized platform for innovation tracking by collaborating with the hackathon team. This initiative fostered a culture of innovation by connecting teams and streamlining the tracking of ideas.

PROJECT

SentimentIQ- AI powered Feedback Intelligence Platform

OPTUM • June 2024 – Present

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AI Capability Dashboard

optum • April 2026 – May 2026

- **Tech Stack:** Node.js, Express.js, TypeScript, MySQL, Kubernetes, Datadog, Claude AI
- Built and delivered a production-grade full-stack AI capability management platform in **15 days** using **Claude AI** for architecture design, code generation, debugging, and test automation.
- Developed scalable REST APIs with Node.js, Express.js, and MySQL powering searchable capability inventory, multi-select filters, pagination, and taxonomy-based exploration.
- Implemented Register/Edit workflows with Microsoft Graph AD integration, field-level validation, and complete audit-trail tracking using timestamped diffs.
- Deployed and monitored the application using Kubernetes and Datadog, ensuring scalable infrastructure and production observability.

- Integrated role-based access controls and approval workflows with Microsoft Graph identities, enabling secure capability updates and clearer ownership across product and engineering teams.

AI-Powered Interior Design Markup Platform

Interior design business • [interior-design-markup-nbyjzsfbu2ndvkwmrmpus.streamlit.app](#) • January 2026 - May 2026

- **Tech Stack:** Python, Streamlit, Generative AI, Computer Vision, Image Processing, Git, Cloud Deployment
- Developed and deployed an AI-powered web application that generates intelligent interior design markups and room transformation workflows from user-provided inputs and images.
- Built an interactive frontend using [Streamlit](#) for real-time design visualization, dynamic user interaction, and rapid prototyping.
- Implemented image-processing and AI-driven rendering pipelines to automate interior layout enhancement and design generation workflows.
- Designed modular backend architecture with reusable components for scalable prompt handling, markup generation, and future AI model integrations.
- Integrated cloud deployment and Git-based version control workflows to support production-ready releases and continuous feature iteration.
- Optimized application responsiveness and usability with low-latency processing and intuitive UI/UX patterns.
- Applied software engineering best practices including clean code structure, modularization, and maintainable project organization.
- Developed prompt templates and image preprocessing checks to convert room photos and client preferences into structured markup suggestions, improving design iteration speed for interior stakeholders.

.open search

Optum • April 2022 - May 2024

- **Tech Stack :** Python, LangChain, Azure OpenAI, Retrieval-Augmented Generation (RAG), Cosmos DB Vector Database, Vector Embeddings, Semantic Search, Machine Learning, NLP, SharePoint Integration, REST APIs, Flask/FastAPI, Azure Cloud, AI Chatbot Development, Document Indexing, Git, Cloud Deployment
- **Developed a search service** transforming data from technical websites, files, and **SharePoint** into a unified **semantic search platform** using **ML techniques** and a **Cosmos vector database**. Indexed over 10,000 files with search times under 5 seconds, saving time for developers and leadership while reducing operational costs.
- **Developed a Chatbot** using **Retrieval-Augmented Generation** and **LangChain** to integrate **Azure OpenAI services**, converting data into embeddings and storing in **Cosmos DB**. Enhanced search response speed by **up to 80%**, significantly improving user interaction efficiency and enabling faster, more accurate query resolution.

EDUCATION

MTech

Indian Institute of Technology Bhubaneswar • Bhubaneswar, Odisha • 2021

- Computer Science and Engineering

BTech

SKILLS

Programming Languages & Frameworks: Java, Python, SQL Server, HTML/CSS, JPA repository, Spring Boot, peewee, JavaScript, Junit, Flask API, Database Design, Machine learning, prompt Engineering, LLM, GPT Models, Prompt Engineering, Generative AI, Open AI GPT-5, RAG, Few-shot prompt, java17 , java 21

Cloud & DevOps: Microsoft Azure, AWS, ECS, GitHub Actions, CI/CD pipelines, Docker, Kubernetes

Web & Database Development: REST APIs, Web Applications, MySQL, HTML5, CSS3, JSON, Cosmos DB

Tools & Platforms: Git, SharePoint, Linux, Windows, Rally, SharePoint Apis, Microsoft Power Automate

Operational Skills: Governance, Time management, Continuous improvement, Requirements analysis, Workflows, Technical support
